



UNIVERSITY OF RUHUNA

Faculty of Engineering

Department of Electrical & Information Engineering

Module Number: EC8204

Module Name: Blockchain and Cyber Security

Project – Group

Common Instructions:

- Two given projects should not be identical. However, it can be in the same domain. In such a case, the application should be different.
- Your presentation name must be named GP_XX_Task_name in short_name.ppt. Eg: GP_01_Gem_Fraud_Detection.ppt
- One submission per group is necessary and sufficient. If you submit multiple presentations per group; the last presentation submitted before the deadline will be considered as the proper submission unless you request by email to the module coordinator.
- Submit to ELMS on or before the **deadline: 31/09/2026**.
- Your presentation should be 3 minutes and presented briefly.
- Maximum 4 members allowed per group.

Task

- Identify a problem where you can use blockchain to provide a solution to. You should implement the proposed solution using any blockchain platform you prefer.
- Please indicate your project selection using the following Google Sheet.

<https://docs.google.com/spreadsheets/d/1eML8xuLvQTjudrwoultCFngpVC9YGHXA3NtJC8vkzDo/edit?usp=sharing>

Some sample project ideas are given below.

- Simple Cryptocurrency Transaction System

Design a basic blockchain to record transactions between users, including block creation, hashing, and chain validation.

- Smart Contract for Student Record Management

Develop an Ethereum smart contract to store and retrieve student academic records securely and immutably.

- Decentralized Voting System

Implement a blockchain-based voting application ensuring transparency, vote integrity, and prevention of double voting.

- Blockchain-Based Certificate Verification System

Create a system to issue and verify digital certificates using blockchain to prevent forgery.

- Decentralized File Integrity Verification

Use blockchain hashes to verify file integrity and detect tampering without storing the actual files on-chain.

- Supply Chain Traceability Using Blockchain

Build a simple supply chain tracking application showing product movement across stages using blockchain transactions.

- Smart Contract for Secure Fund Escrow

Implement a smart contract that holds funds in escrow and releases them automatically based on predefined conditions.

- Blockchain-Based Identity Management System

Design a basic decentralized identity (DID) system allowing users to manage and prove their identity attributes.

- Blockchain Security Analysis of a Public Network

Analyze common attacks (e.g., 51% attack, smart contract vulnerabilities) on an existing blockchain and propose mitigation strategies.

- Decentralized Bug Bounty or Reporting Platform

Create a simple blockchain-based platform where security vulnerabilities can be reported and rewarded transparently.